

Question 2: AI HW

a) for minimization of simulated annealing: $f(x, y)$

$$\Delta = f_{\text{candidate}} - f_{\text{current}}$$

if $\Delta \leq 0$, we accept otherwise we accept with probability

$$P = e^{-\Delta/T}$$

Booth: $f(x, y) = (x + 2y - 7)^2 + (2x + y - 5)^2$
 $-10 \leq x, y \leq 10$

$$f(0, 0) = (0 + 0 - 7)^2 + (0 + 0 - 5)^2$$

$$= 49 + 25 = 74$$

T: 1 start value

K = 100; iterations

~~Booth~~ =

N.S = 0.5

temp decay = 0.1

Neighborhood size = 0.5 (max one movement size)

Initial temp T = 1

K 100 (iteration before temp change)

Temp decay 0.1 Amount ignore after 100 iterations

Booth bounds [-10, 10]

Himmelblau bound [-5, 5]

Griewank bound ~~[-30, 30]~~ (-30, 30)

some random val

eg $-10 \leq x, y \leq 10$

$(5, -2), (-7, 8), (0.6, 3.2)$

own value (test) $f(0,0) = 74$

neighborhood size = 0.5

$\Delta x \in [-0.5, 0.5]$

$\Delta y \in [-0.5, 0.5]$

eg $\Delta x = 0.3$

$\Delta y = 0.4$

$x_{\text{candidate}} = x + \Delta x \Rightarrow 0 + 0.3 = 0.3$

$y_{\text{candidate}} = y + \Delta y = 0 + 0.4 = 0.4$

$(0.3, 0.4) \rightarrow$ in bound $-10 \leq x, y \leq 10$

candidate value:

$f(0.3, 0.4) = 50.81$

$50.81 < 74 \checkmark$

one iteration $\checkmark$ accept

out of 3

Pseudocode:

function SA (prob-function, initial-T, cooling-rate, iterations-per-T):
 current-state = random-state-in-bound (bound)
 current-value = problem-function (current-state)

 best-value = current-value

 best-state = current-state

 T = initial-T

 while T > 0:

 for i from 1 to iterations-per-T:

 next-state = generate-random-neighbor (current-state)

 next-value = problem-function (next-state)

 delta-E = next-value - current-value

 if $\text{delta-E} < 0$ or $\text{Random}(0,1) < \exp(-\text{delta-E}/T)$:

 current-state = next-state

 current-value = next-value

 if current-value < best-value:

 best-state = current-state

 best-value = current-value

 T = T - cooling-rate

return best-state, best-value